

BIOTECHNOLOGY

PRINCIPLES AND PROCESSES



WHAT IS BIOTECHNOLOGY?

- Biotechnology is the use of living organisms, cells or their components to develop useful products & processes.
- It combines biology with technology to solve problems & improve human life.

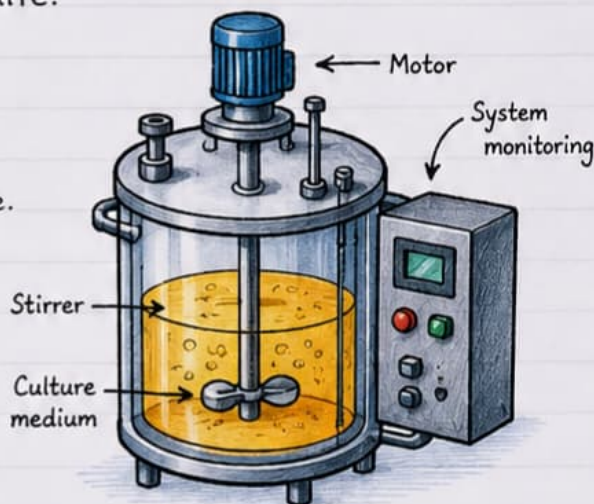
KEY IDEA



Using life to improve life. ♥

IMPORTANT TERMS

- ♥ **Organism** : A living individual.
- ♥ **Cell** : Basic structural & functional unit of life.
- ♥ **Bioprocess** : Use of biological systems or organisms to make or modify products.
- ♥ **Bioreactor** : A vessel in which biological reactions are carried out under controlled conditions.



Simple Bioreactor

PRINCIPLES OF BIOTECHNOLOGY

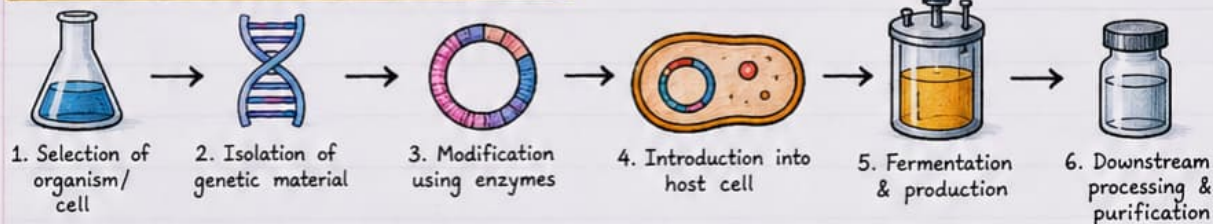
- ① Genetic material (DNA/RNA) is the basis of biotechnology.
- ② DNA can be manipulated using enzymes.
- ③ Foreign DNA can be introduced into host cells for desired results.
- ④ Microorganisms and their products are used in various processes.

EXAMPLES

- ★ Insulin from bacteria
- ★ Vaccines from weakened microbes
- ★ Bt crops for pest resistance



MAJOR STEPS IN BIOTECH PROCESSES



APPLICATIONS (OVERVIEW)

- **Medicine** : Insulin, vaccines, antibiotics
- **Agriculture** : Bt crops, pest resistance
- **Environment** : Sewage treatment, bioremediation
- **Industry** : Enzymes, biofuels, organic acids

Biotechnology is not just a subject, it's a revolution in modern science.



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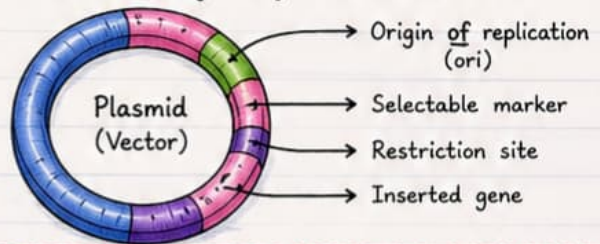


RECOMBINANT DNA TECHNOLOGY (r-DNA TECHNOLOGY)

- It involves manipulation of DNA to isolate a gene of interest and introduce it into a host organism to **express** and **produce** the desired product.
- Steps include isolation of genetic material, cutting with **restriction enzymes**, ligation into a **vector**, transfer to host and expression.

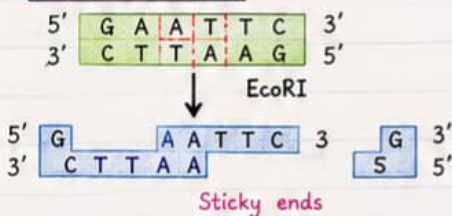
TOOLS OF r-DNA TECHNOLOGY

- ① **Cloning Vector** - A small DNA molecule that can carry foreign DNA into host cell.

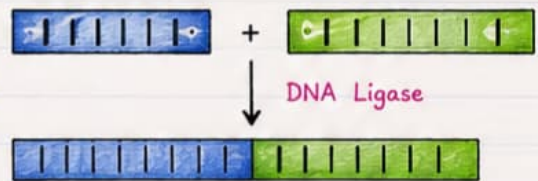


- ② **Restriction Enzymes** - Act as molecular scissors and cut DNA at specific sequences.

Example : EcoRI



- ③ **DNA Ligase** - Joins DNA fragments together using ATP.



PROCESS : r-DNA TECHNOLOGY (Flow)

1. **Isolation of DNA** - Extract DNA from source organism.
2. **Cutting of DNA** - Use restriction enzymes to cut gene of interest & vector.
3. **Ligation** - Join gene with vector using DNA ligase → recombinant DNA.
4. **Transformation** - Introduce r-DNA into host cell (e.g., bacteria) by heat shock or electroporation.
5. **Selection** - Use selectable marker to identify transformed cells.
6. **Cloning & Expression** - Host cell multiplies and expresses the gene.
7. **Downstream Processing** - Recovery and purification of the desired product.

TYPES OF BIOTECHNOLOGY

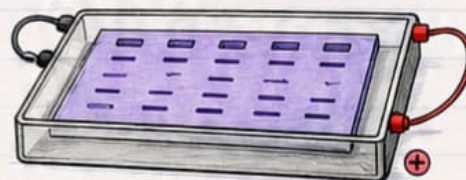
- **Green Biotechnology** - Agriculture, plants, crop improvement.
- **Red Biotechnology** - Medical, pharmaceuticals, vaccines, gene therapy.
- **White Biotechnology** - Industrial processes, enzymes, biofuels.
- **Grey Biotechnology** - Environmental protection, waste treatment, bioremediation.
- **Blue Biotechnology** - Marine and aquatic resources.

KEY TAKEAWAY

💡 r-DNA technology enables production of valuable products like insulin, vaccines, enzymes & many more on a large scale.

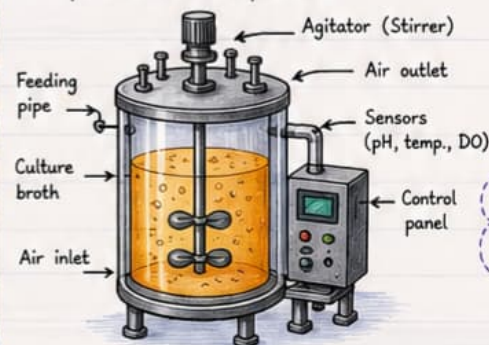
TECHNIQUES USED

1. **Gel Electrophoresis** - Separates DNA fragments based on size.



Result
↓
Smaller fragments move further towards +ve electrode.

2. **Fermentation Technology** - Large scale culturing of microorganisms in bioreactors to produce desired product.



Maintains optimum conditions for maximum yield.

THINK & REMEMBER

- Why are restriction enzymes called molecular scissors?
- What is the role of a selectable marker?
- Give examples of products made using biotechnology.



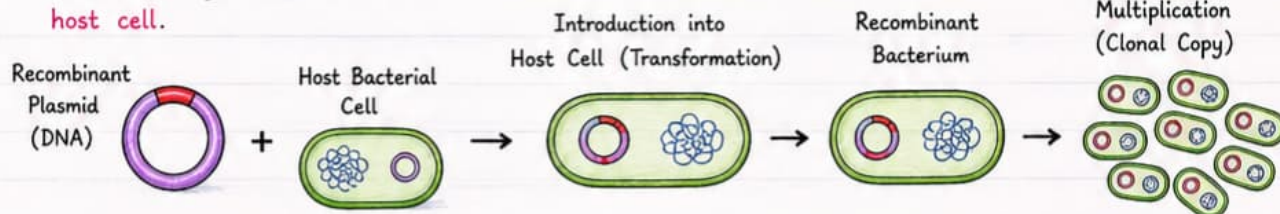
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RECOMBINANT DNA TECHNOLOGY (Contd.)

→ Insertion of gene of interest into the **host cell**.

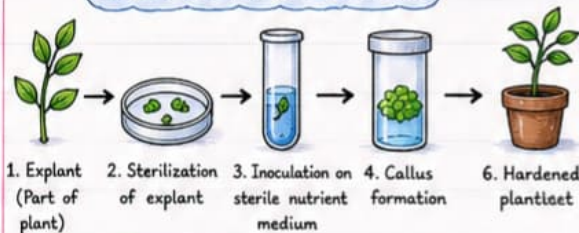


★ **Application** : Production of human insulin, growth hormone, interferon, vaccines, etc.

PLANT BIOTECHNOLOGY

- Based on the **totipotency** of plant cells to develop into a whole plant.
- Involves tissue culture techniques.

STEPS IN TISSUE CULTURE



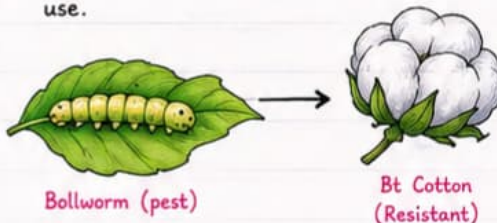
APPLICATIONS

- Propagation of disease free plants.
- Rapid multiplication of elite plants.
- Production of plants in all seasons.
- Conservation of endangered plants.



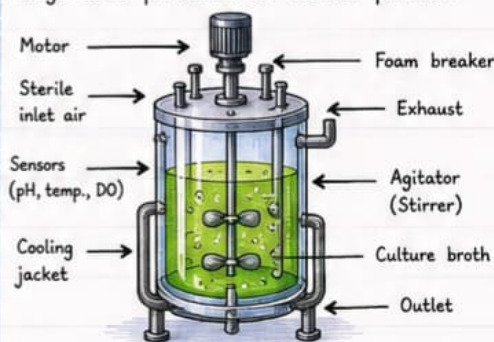
EXAMPLE - Bt COTTON

- Gene (**cry gene**) from *Bacillus thuringiensis* coding for insecticidal protein is isolated.
- Inserted into cotton plant genome.
- Transgenic cotton produces Bt toxin which is toxic to bollworms.
- Result → Increased yield, less pesticide use.



BIOREACTOR - THE HEART OF BIOTECH PROCESS

A bioreactor provides optimum conditions for large scale production of desired product.



CONTROLLED CONDITIONS

- ✓ Temperature
- ✓ pH
- ✓ Dissolved oxygen (DO)
- ✓ Nutrients
- ✓ Sterility

PRODUCTS FROM MICROBIAL FERMENTATION

Product	Microorganism	Use
Antibiotics	Penicillium, Streptomyces	Treatment of bacterial infections
Vitamins	Lactobacillus	Nutritional supplements
Organic acids	Aspergillus, Lactobacillus	Food, beverages, chemicals
Enzymes	Bacillus	Detergents, food, pharma
Alcohol	Saccharomyces cerevisiae	Beverages, biofuel
Single cell protein	Spirulina	Animal feed, food

DOWNSTREAM PROCESSING

Recovery and purification of the desired product from the culture.

- ★ Separation (filtration, centrifugation)
- ★ Purification (chromatography, precipitation)
- ★ Formulation (conversion to stable form)



QUICK RECAP

- ✓ Biotechnology = Biology + Technology.
- ✓ r-DNA technology allows manipulation of genes.
- ✓ Microorganisms are used for large scale production.
- ✓ Plant tissue culture helps in rapid propagation.
- ✓ Biotechnology has vast applications in medicine, agriculture, industry and environment.

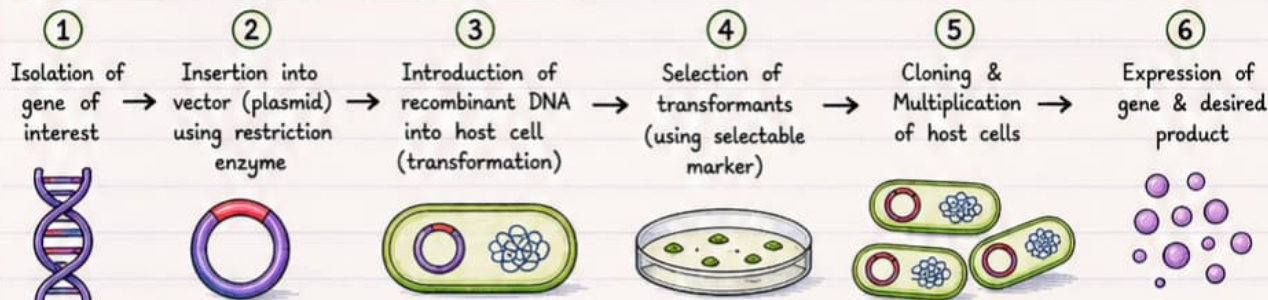


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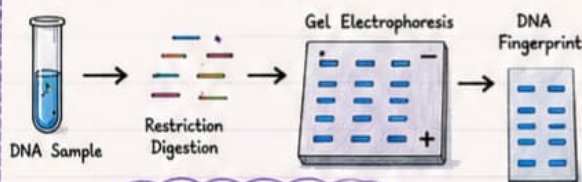


GENE CLONING - KEY STEPS (SUMMARY)



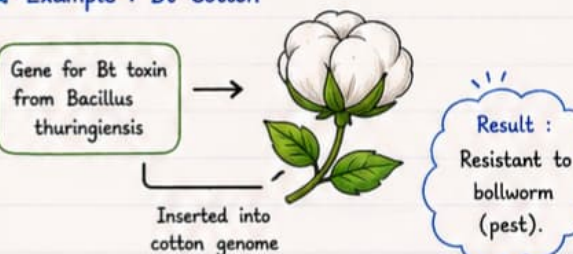
DNA FINGERPRINTING

- ★ Technique to find variations in repeated sequences (VNTRs) in DNA.
- ★ Each individual has a unique pattern.
- ★ Used in forensic science, paternity testing and identification of individuals.



TRANSGENIC ORGANISMS

- ★ Organisms that carry foreign gene (transgene) in their genome.
- ★ Made using genetic engineering.
- ★ Example : Bt Cotton



BIOTECHNOLOGY IN MEDICINE

- ♥ Production of recombinant insulin, human growth hormone, interferons, vaccines, clotting factors.
- ♥ Gene therapy - To treat genetic disorders.
- ♥ Monoclonal antibodies - Used in diagnosis & therapy (cancer, autoimmune diseases).
- ♥ Stem cells & tissue engineering - Regenerative medicine.



ETHICAL & ENVIRONMENTAL CONCERNS

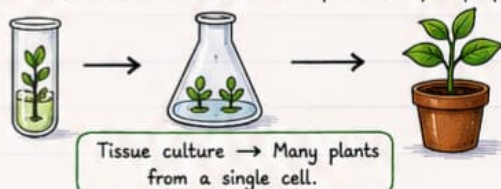
- Release of GMOs may affect biodiversity.
- Gene flow from GM crops to wild relatives.
- Allergic reactions & health issues.
- Animal rights & bioethics.
- Need for proper testing and regulations.



Biotechnology must be used responsibly for the welfare of all living beings and the environment. ♥

BIOTECHNOLOGY IN AGRICULTURE

- ★ Bt crops - Resistant to pests (e.g., Bt cotton).
- ★ Herbicide tolerant crops - Survive in presence of herbicides.
- ★ Improved yield, resistance to diseases, stress tolerance (drought, salinity).
- ★ Tissue culture - Disease free plants, rapid propagation.



FUTURE PROSPECTS

- ✓ CRISPR-Cas9 and genome editing - Precise changes in genes.
- ✓ Synthetic biology - Designing new biological systems.
- ✓ Bioinformatics - Helps in analyzing huge biological data.
- ✓ Personalized medicine - Treatment based on individual genetic makeup.
- ✓ Sustainable solutions for food, health, energy and environment.



BIOTECHNOLOGY = INNOVATION + RESPONSIBILITY

Let's use it wisely for a better and healthier world. 🌍 ♥

